

Configuration and Uncertainty Sensitivity of DP-SGD Utility Recommendations: An ACS Decision Audit

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Abstract

Utility-constrained recommendations for DP-SGD depend on training configuration, validation uncertainty and the environments used for selection. We audit these decisions using matched ordinary, clipping-only and noisy controls, frozen validation choices and held-out ACS state tests. A synthetic study and ACS 2018 pilot motivate an independently frozen ACS 2017 replication with 840 fits across two representations. Joint configuration search produces successful recommendations in all 20 seed cases per representation, compared with zero or one for a restricted epsilon-only baseline. Source-only joint search matches shift-aware search on this replication, so expansion of the candidate bank explains the availability gain. A supplementary 400-fit comparison evaluates published AUTO-S clipping under a fixed matched recipe. It produces no uncertainty-bound recommendations, while standard norm search retains high yield; this is not a tuned algorithm ranking. Fresh simultaneous calibration reproduces undercoverage in a small rare-label clustered scenario (86.9% against nominal 95%). A conservative concentration sensitivity covers every valid Gaussian simulation but abstains on all ACS cases. These results support configuration-sensitive decision auditing and explicit coverage/abstention reporting, without a new mechanism, reliable ACS certification or established general benefit of shift-aware selection.

1. Research question and relationship to the MSc

The original MSc compared privacy-related perturbations on a small public insurance table, including smoker prediction with a strong charges proxy. The revised benchmark introduced accounted DP-SGD, leakage-safe evaluation and empirical privacy auditing. The present study remains adjacent to that work through tabular prediction, privacy-utility trade-offs and honest measurement, but asks a narrower decision question:

When a validation rule recommends a DP-SGD budget subject to a relative utility-loss margin and an absolute AUC floor, which recommendations survive held-out distribution shift, and how sensitive is the answer to clipping, training configuration and validation uncertainty?

This study concerns utility reliability. It does not infer membership privacy from utility, equate a failed attack with absence of leakage, or revive the unsupported sex-specific leakage hypothesis. The earlier LiRA/power work remains a separate study. Insurance motivates the problem; controlled synthetic data and ACS provide the current evidence.

2. Closest prior work and bounded contribution

Mahajan, Tople and Sharma [1] connect stable representations, distribution generalization and membership inference, including utility differences at the same formal DP guarantee. Therefore, discovering that stable features can improve private utility is not a defensible novelty claim here.

Zhang, Pang and Mauw [2] study privacy disparities associated with spurious groups and the distinction between prediction robustness and memorization. Our utility recommendation evaluation does not reproduce their privacy conclusions or claim to introduce spurious-feature privacy analysis.

DP-UTIL [3] compares utility and privacy across perturbation locations. Bagdasaryan and Shmatikov [4] explain disparate accuracy effects of clipping and noise. Consequently, broad mechanism comparisons and the existence of clipping-related utility loss are established background.

The candidate incremental contribution is a reproducible **decision audit**: frozen validation-only budget choices; explicit abstention and two failure criteria; matched clipping/noise controls; paired clustered uncertainty; development separated from a prospective synthetic primary comparison; and external state tests. None of these ingredients alone is new. Their combination may support a useful empirical benchmark, but a combination is not itself proof of publication-level novelty. The results should be positioned as recipe sensitivity and negative evidence against broad stress-test claims.

A subsequent targeted survey adds important constraints. Morsbach et al. [8] already establish joint clipping/learning-rate effects. Panda et al. [9] combine private HPO with OOD evaluation. DomainBed [10] treats model selection as essential to domain generalization. Accuracy First [11] and Brownian Noise Reduction [12] already optimize privacy subject to accuracy requirements. Their methods analyze the private release and adaptive utility checks; our fixed public-data bank does not implement those mechanisms. Papernot and Steinke [16] and Koskela and Kulkarni [17] likewise account for search privacy, which is outside our per-candidate accounting. The supplementary training comparator uses published AUTO-S [14], not a new clipping method. Consequently, neither the selection problem nor a joint search is our method novelty. The full survey maps sixteen related sources and the remaining evidence boundary.

3. Methods

3.1 Models and training controls

The study uses a two-layer MLP (16 hidden units, tanh, one binary logit), cross-entropy and SGD. All controls use matched initialization and Poisson sampling streams with a common expected-batch gradient normalization. The three conditions are ordinary training, per-record clipped training with zero noise, and DP-SGD at requested ϵ values 2 or 8. Noise uses a separate, fixed reproducible random stream. The RDP accountant records achieved ϵ , $\delta=10^{-5}$, sample rate, noise multiplier, steps and empty batches per model.

Development uses 3,000 training rows, independent 2,000-row validation/test cohorts, batch 256, 50 epochs and learning rate 0.1. Synthetic development crosses strengths 0.75/0.90, clipping norms 0.5/1/2/5 and five seeds 10-14. ACS crosses norms 1/5 and the same five seeds. These account for 160 and 40 model fits respectively, including ordinary controls repeated across norms. Those repeated controls are not independent observations.

This is a controlled recipe comparison, not a claim that each configuration has been independently optimized. No alternative architecture benchmark or convergence guarantee is established. Raising the norm also raises absolute noise for a given noise multiplier, so the contrast is a training-configuration effect rather than an isolated causal effect of clipping.

3.2 Environments and data

The synthetic generator and all parameters are available in `src/dp/proxy_shift.py` and the JSON protocols. Train, validation and test streams are independent within seed. Validation proxy retention is 0.5/0.75; test retention is 0/0.25. Generic perturbations provide a comparison validation rule. Proxy intervention is controlled by construction, not inferred from real-world correlations.

The external pilot downloads official 2018 one-year ACS person archives. California supplies source train/validation/test partitions; Oregon and Washington supply natural-shift validation; Nevada and Arizona supply external tests. ACSIncome eligibility is age >16, income >100, working hours >0 and person weight ≥ 1 , with binary target income >\$50,000 [5,6]. Classification is unweighted and concerns public sampled records, not survey-weighted population income estimates.

Encoding uses fixed public bounds for numeric features, fixed categorical domains and deterministic modulo hashing for occupation and birthplace. It differs from the standard Folktables representation. Households are allocated wholly to distinct source partitions; the last household in a partition can be truncated, with its unused records excluded from subsequent partitions. Occupation-block zero masking is a hypothetical validation stress test; it is not a causal explanation of state differences. State shifts can change multiple variables and relationships simultaneously.

ACS training seeds resample from the same fixed state archives, so they are algorithm/cohort replications rather than independent state replications. The five states and one year restrict external validity.

3.3 Recommendations and held-out evaluation

For each configuration the recommendation chooses the smallest achieved ε among candidate budgets satisfying both AUC loss ≤ 0.03 relative to the ordinary reference and private AUC ≥ 0.70 in every specified validation environment. Failure to meet these constraints yields abstention. Source-only, proxy-stress and generic/natural-shift rules are compared using point estimates or uncertainty bounds. Privacy caps are fixed by the protocols, not justified as acceptable deployment privacy levels.

All choices are persisted before external test scoring; ACS external test records are loaded after choices are written. Test failure occurs when either the relative gap exceeds 0.03 or private AUC falls below 0.70. Counts are reported alongside abstentions. Evaluations in two test environments share a model and cannot be treated as independent Bernoulli trials. Increasing abstention can reduce selected-case failures without improving budget choice.

3.4 Paired uncertainty and decomposition

AUC structural influences preserve ties and covariance between predictions on the same labeled cohort. Variance sums record influences within households for ACS, then uses a cluster sandwich estimate; synthetic records form their own clusters. One-sided t bounds use a Bonferroni family across validation environments, candidate budgets, clipping norms and both gap/floor endpoints. At least 20 records per class and 30 clusters are required. Boundary AUCs with degenerate variance are marked unsupported rather than certified.

These are asymptotic, conditional-on-model bounds for sampled validation environments. They do not certify unseen-state utility or constitute a new confidence procedure. Cluster independence is an approximation for ACS and does not model all survey design dependence. A Gaussian clustered simulation and a paired household bootstrap provide limited implementation/calibration checks; the broader simultaneous check in Section 4.5 finds undercoverage in one small imbalanced scenario and does not validate ACS survey-design coverage.

Let A^o , A^c , A^d denote ordinary, clipped and DP AUC. The relative gap decomposes exactly as $(A^o - A^d) = (A^o - A^c) + (A^c - A^d)$. The corresponding shift interaction subtracts the source gap from the shifted gap. Components are matched training comparisons, not a general causal mediation analysis; added-noise components can be negative.

3.5 Prospective primary comparison

After development, the protocol fixed one primary contrast: norm-1 minus norm-5 **total gap interaction**, at $\varepsilon=2$, strength 0.90 and test retention 0. The development paired SD was 0.0087875 AUC. Normal-approximation planning used a 0.01 effect, two-sided $\alpha=0.05$ and target power 0.80, with a minimum 20 fresh seeds. The protocol uses seeds 100-119, 4,000 validation/test rows and unchanged training settings. It was committed as `1feb9fe` before that run started. This is a locally prospective protocol, not an independently timestamped public preregistration. Five-seed variance planning is uncertain.

The primary estimate uses the paired seed mean and Student-t 95% interval. All other confirmation-run summaries are descriptive. A non-significant test does not imply equivalence; practical interpretation must use the interval and the specified 0.01 margin. The tested population is the specified synthetic generator/recipe, not arbitrary tabular tasks.

4. Results

4.1 Synthetic development

At norm 0.5 and $\varepsilon=2$, mean complete-proxy-removal interaction was approximately 0.02994 AUC, consisting of 0.02964 clipping-associated and 0.00030 added-noise components (pooled over both development strengths for description only). At norm 1 the corresponding mean was approximately 0.00093; at norm 5 it was approximately -0.00027. These changes challenge a recipe-independent clipping-dominance story. They also show why further mechanistic claims need stronger comparisons than a single low-norm pilot.

Source-only selection failed in 12/20 held-out evaluations at norm 0.5 and 5/20 at norms 1, 2 and 5. Proxy stress at norm 0.5 selected seven of ten seed/strength cases and failed in 6/14 evaluations; all higher-norm rules selected ten cases and failed in 5/20. Point and bounded decisions were equal in this development run.

These counts do not demonstrate that proxy stress generally improves transport. Absolute-floor violations can remain when the ordinary reference also loses signal under shift.

4.2 ACS external pilot

At $\varepsilon=2$, ordinary/private source-test mean AUCs were 0.85578/0.82488 for norm 1 and 0.85578/0.84627 for norm 5. Norm-1 mean clipped-only gradient clipping rate was approximately 54.9%, versus 1.69% for norm 5. Under norm 1, point source-only and natural-shift rules each selected three seeds and failed in one of six external evaluations; proxy stress selected two and failed in one of four. Bounded selection abstained on all five seeds under every norm-1 rule.

Under norm 5, every rule and bound choice selected all five seeds and all ten external evaluations passed. This is a pilot observation, not a population failure-rate bound. Mean private AUC at $\varepsilon=2$ was 0.82875 in AZ and 0.81291 in NV; the corresponding ordinary means were 0.84276 and 0.82428. The larger norm reduces clipping-related source utility loss here while leaving some added-noise loss. The external relative gaps do not uniformly amplify source gaps; natural shift is not equivalent to synthetic proxy removal.

4.3 Interval checks

Across 400 Gaussian clustered simulations, marginal one-sided 95% coverage was 0.950 for the gap upper bound and 0.960 for the AUC lower bound. Nominal Monte Carlo standard error is 0.01090. This single DGP does not validate ACS survey dependence, tail cases or simultaneous selection coverage.

For the predetermined ACS source-validation seed-10, norm-1, $\varepsilon=2$ cell, the paired influence gap upper bound was 0.03419 and the 999-replicate paired household percentile-bootstrap bound was 0.03436. Candidate AUC lower bounds were 0.82184 and 0.82052. Both marginal gap bounds exceed the 0.03 tolerance. This is a sensitivity check using two diagnostic retrains, not additional independent experimental evidence or the multiplicity-adjusted selection rule.

4.4 Fresh-seed primary result

The exact primary estimate and interval are recorded in the confirmation stage's `primary_result.json`. Across 20 fresh seeds, the norm-1 minus norm-5 interaction was **0.008275 AUC**, with paired-mean 95% interval **[0.004092, 0.012458]** and two-sided $p=0.000556$. The positive contrast is distinguishable from zero in the specified experiment. Its interval straddles the planned 0.01 practical-effect margin; it establishes neither an effect above that margin nor equivalence within that margin.

Descriptively, every rule selected all 20 seeds under both norms. Norm 1 failed in 20/40 held-out evaluations and norm 5 in 19/40; proxy stress and bounded selection made the same choices as source-only selection. These paired environment counts are not the primary statistical test and do not establish a recommendation-policy improvement. There were 120 confirmation fits, including repeated ordinary controls.

4.5 Independent ACS 2017 configuration-selection extension

Before external scoring, a separate protocol fixed CA source, OR/WA validation and previously unused CO/UT external tests. Twenty fresh seeds 200-219 were run for each of the original hashed representation and a public decimal-digit one-hot sensitivity. Three recipes (16-unit MLP/50 epochs/LR .1, 32-unit MLP/100 epochs/LR .05, logistic/100 epochs/LR .1), norms 1/5 and epsilon 2/8 provide 420 fits per encoding, **840 in total**. Cohort sizes, batch size and utility limits remain 3000/2000, 256, .03 gap and .70 floor. The common ordinary reference is selected using source-validation AUC only. All decisions and the reference are persisted before loading external tests. The confidence family also covers every possible ordinary reference choice.

The primary endpoint is successful recommendation yield over all seed/cohort cases; a selected case must satisfy both utility constraints in both external states. Abstention is zero successful yield. At zero safety margin, bounded fixed-recipe epsilon/source selection succeeds in 0/20 hashed and 1/20 digit cases, versus 20/20 for joint-recipe/shift search. **Source-only joint clipping and source-only joint recipe search also succeed in 20/20 under both encodings.** The primary yield contrast is therefore +1.00 and +.95, but demonstrates no additional success-yield benefit from shift-aware validation over competent source-only configuration search. Search sizes differ; this is not a claim of HPO efficiency or new selection methodology.

The original specified paired t summary is retained, with its degenerate hashed interval flagged unsupported. A dated supplementary conservative paired binary construction uses Bonferroni-combined Clopper-Pearson

discordance intervals: $[.6065, 1.0000]$ and $[.5239, .9994]$. These intervals concern independent seed streams on a fixed split, not independent states or universal transport. The supplementary estimator is standard and does not fix AUC-bound coverage.

The point fixed-recipe source policy succeeds in 10/20 hashed cases (ten abstentions), and 17/20 digit cases (one abstention, two selected failures). Representation changes recommendation behavior. Bounded logistic-only shift selection also succeeds in all 20 cases under each encoding. Both source-only and shift-aware joint policies continue to succeed in 20/20 at safety margin .01. All predefined margin curves and exact-coverage comparisons are retained; no threshold is optimized against test failures.

For the hashed base MLP at epsilon 2, mean source clipping/noise gaps are .02884/.00136 at norm 1, versus .00010/.01153 at norm 5. This illustrates a bias/noise balance under the same accounted privacy target. The stronger MLP has similar ordinary utility, and does not eliminate configuration sensitivity. These descriptive components are not a general causal mechanism claim.

The full-family diagnostic has 600 clustered Gaussian simulations with 12 candidates, five environments and three possible references (360 endpoints). Observed simultaneous coverage is .880/.945/.980 in three predefined imbalance/cluster scenarios; Monte Carlo SEs are .0230/.0161/.0099. The .880 result limits any claim of uniform nominal .95 calibration. The **bounds_supported** flag checks computational nondegeneracy, not coverage. The study uses explicitly asymptotic estimates and does not provide certified utility. Detailed findings and provenance retain the complete protocols, launch hashes, archive checksums and accounting.

4.6 Supplementary published-clipping comparison

After inspecting the original replication and calibration outcomes, we froze a dated supplementary protocol and used new seeds 300-319 on the same known ACS 2017 split. This is supplementary evidence, not a new independent-domain confirmation. Each representation adds 200 fits: one ordinary reference, three transformed/noise-free controls and six DP candidates per seed. AUTO-S [14] uses all-parameter per-example normalization $g_i/(\|g_i\|_2 + .01)$, unit sensitivity and the same Opacus noise, expected-batch normalization and accounting. Standard C=1/C=5 and AUTO-S share the base 16-unit MLP, 50 epochs and LR .1. These fixed choices are not tuned after test outcomes and do not reproduce the published vision/NLP suite.

All rules have the same 36-endpoint validation family and fixed ordinary reference. This differs from Section 4.5’s 360-endpoint bank/reference family; cross-study yield changes do not isolate an algorithm effect. Within this supplement, successful cases require both CO and UT to meet the original gap .03 and AUC floor .70. The following rows use safety margin zero:

Representation	Selection / uncertainty	Selected	Successful
Hashed	AUTO-S source / point	0/20	0/20
Hashed	AUTO-S source / sandwich	0/20	0/20
Hashed	Standard joint source / point or sandwich	20/20	20/20
Hashed	Combined shift / point or sandwich	20/20	20/20
Digits	AUTO-S source / point	10/20	5/20
Digits	AUTO-S shift / point	2/20	2/20
Digits	AUTO-S source or shift / sandwich	0/20	0/20
Digits	Standard joint source / point or sandwich	20/20	19/20
Digits	Combined shift / point or sandwich	20/20	20/20
Both	Every policy / concentration sensitivity	0/20	0/20

AUTO-S does not improve recommendation yield under this one schedule. This does not establish inferiority after appropriate learning-rate/epoch tuning. Combined shift versus combined source has yield difference 0 for hashing and .05 for digits. Conservative paired seed-case intervals are $[-.1968, .1968]$ and $[-.1961, .2793]$, respectively. The digit gain is one case; the interval includes zero and does not establish a reliable shift-validation benefit. These are supplementary descriptive contrasts, not multiplicity-adjusted confirmatory tests. Both standard source and shift joint rules had 20/20 success in the original independent replication; retain both findings.

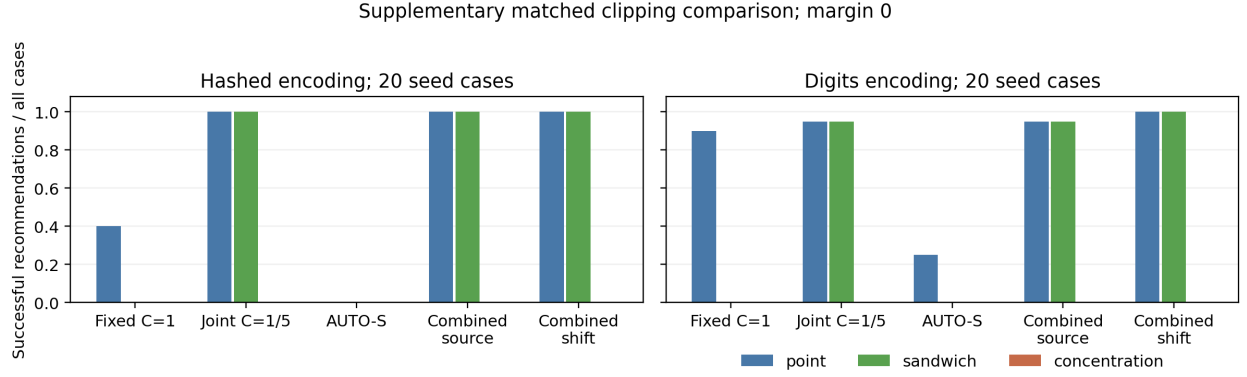


Figure 1: Successful recommendation yield for the supplementary matched clipping study. The conservative sensitivity abstains for every policy.

4.7 Fresh calibration and a conservative sensitivity

Fresh simulation seed 20261002 repeats the three known Gaussian clustered scenarios with 200 draws each and the original 360 endpoints. One rare-label draw has fewer than 20 records/class and is excluded by the original validity rule; 599 draws are valid. The original sandwich procedure is unchanged.

Prevalence / households	Sandwich simultaneous coverage	Exact binomial 95% interval	Concentration coverage	Mean raw gap radius: sandwich / concentration
.10 / 100	173/199 (.8693)	[.8144,.9128]	199/199	.1404 / 1.1472
.30 / 250	191/200 (.9550)	[.9163,.9792]	200/200	.0653 / .5901
.50 / 250	194/200 (.9700)	[.9358,.9889]	200/200	.0486 / .5704

The rare-label failure reproduces the original .880 coverage result. It cannot be repaired by relabelling a nondegenerate computation as supported inference. The manuscript therefore retains sandwich selection as an asymptotic heuristic and removes dependable certification from its contribution claim.

For a conservative sensitivity, apply standard bounded differences [15] to independent score-generating households conditional on the label vector and cluster membership. With P/N positive/negative records and household counts p_h, n_h , changing a household’s scores affects at most $c_h = p_h/P + n_h/N - p_h n_h / (PN)$ of all positive-negative rank pairs. Let $r = \sqrt{\frac{1}{2} \sum_h c_h^2 \log(M/\alpha)}$ and $w = \sum_h p_h n_h / (PN)$, where M is the endpoint family. A candidate-AUC kernel has range $[0,1]$; a paired-gap kernel has range $[-1,1]$. Conditional-expectation radii are r and $2r$. Under a common class-conditional marginal score law across households **after conditioning on the entire label vector**, within-household pair bias relative to the marginal target is bounded by w and $2w$. Use candidate lower AUC $\hat{A} - r - w$ and upper gap $\hat{G} + 2r + 2w$.

These conditions hold in the simulated score DGP, but need not hold in ACS: households may be dependent and conditional score laws may vary with their labels and other attributes. Thus concentration selection on ACS is an assumption-dependent sensitivity, not a survey-design or unseen-state certificate. All 599 valid simulations are covered; the coverage lower confidence limits are about .982, not proof of universal coverage. The bounds are so wide that every ACS rule abstains, even at zero safety margin. Exact seed-yield intervals do not remedy AUC uncertainty or survey assumptions.

5. Interpretation and submission boundary

The evidence currently supports checking clipping/training configuration before telling a privacy-noise or proxy-robustness story, and reporting abstention alongside selected-case failures. It does not establish a new DP protection phenomenon or universally superior recommendation policy. Related work already establishes the broad stable-feature and clipping/noise issues. The defensible publication direction is a narrow reproducible benchmark/negative-result contribution, subject to a reviewer finding the decision evaluation increment useful.

The independently frozen ACS replication, representation sensitivity and stronger recipe/logistic comparison have now been executed (Section 4.5). They narrow the interpretation to configuration-sensitive recommendation availability: shift-aware search adds no successful-yield improvement on the examined splits. Before

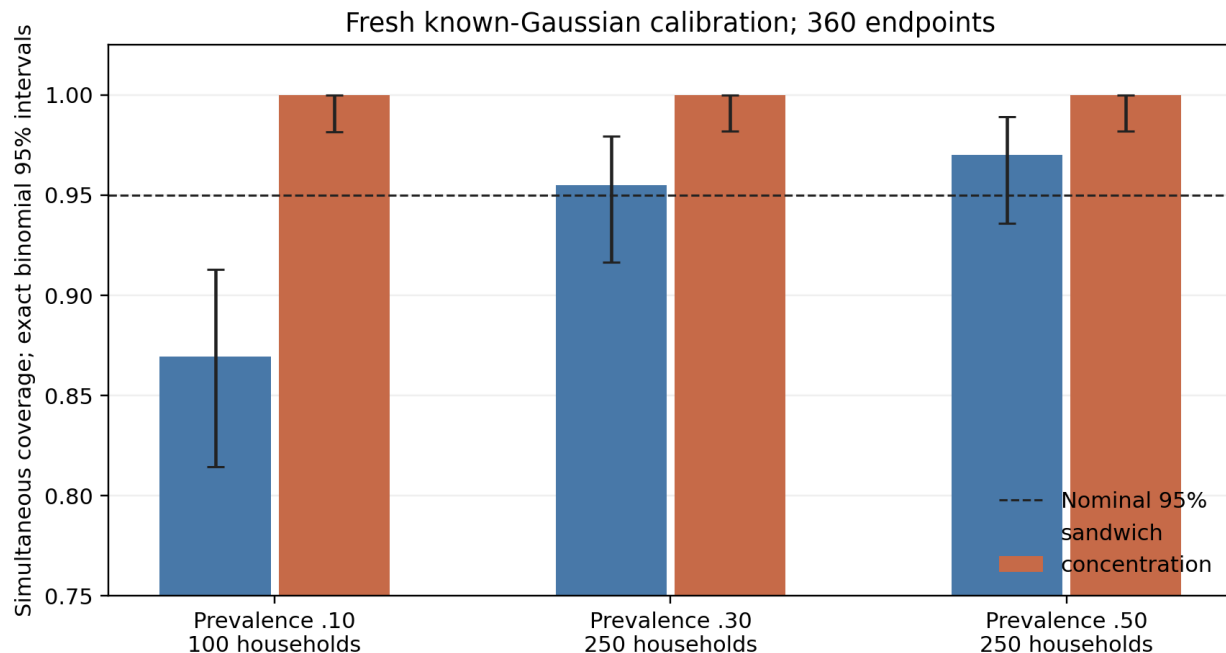


Figure 2: Fresh complete-family calibration. Error bars describe Monte Carlo coverage uncertainty; they are not ACS validation intervals.

interest, without establishing acceptance. Author and external expert review remain necessary before an actual submission.

6. Privacy scope, reproducibility and limitations

The Opacus accountant covers each prepared-record DP-SGD model under the implemented Poisson sampling and add/remove adjacency. It does not account for the many-model research release, public evaluation/selection or the older insurance preprocessing. Fixed public ACS encoding avoids learning feature domains from private records, but household-safe partitions and household-cluster utility inference do not provide household DP. The study uses public data and reproducible non-cryptographic randomness; it is not a production privacy deployment or an acceptable- ϵ policy.

Configs, per-model accounting, frozen choices, metrics, decisions, archive checksums and calibration results are committed. Raw ACS archives and prepared records are excluded from Git. Runtime/provenance notes identify executed revisions, including the development runner’s stage-label metadata change during execution. Reported source hashes captured at run completion must not be mistaken for an immutable launch snapshot.

Commands and the prospective protocol are in `docs/PUBLICATION_PLAN.md`. Reproduction requires a fresh output directory, the listed CPU runtime and official data access. Seeds reproduce experimental streams conditional on software/hardware; tests check invariants rather than certify scientific conclusions. The historical PDF and unsupported demographic significance claims must not be used as the submission manuscript.

The supplementary protocol was committed locally before launch; its identical tree was pushed during execution. Source hashes verify no training-code change between launch and completed manifests. It is not independently registered, and the external state/year outcomes were already known. A source-equivalent public protocol snapshot is commit `9fe78890ed716b2f4ec808d7f3389d326acc203a`. The new run-verification checks account for all 400 fits, frozen choices and their subsequent decisions. They validate saved invariants, not a second full reproduction or the statistical assumptions.

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